THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** biodiversity	Diversity of life.
2	**(a)** algae	Thallophyta.
3	**(b)** Porifera	Pores on the body.
4	**(b)** gills	Aquatic.
5	**(d)** taxonomy	Naming and grouping.
6	**(b)** bryophytes	Need water for fertilisation.
7	**(d)** Cnidaria (Coelenterata)	Stinging cells.
8	**(b)** Amphibia	Live on land and water.
9	**(c)** Carolus Linnaeus	Two-name system.
10	**(d)** pteridophytes	Ferns.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Variety of living organisms on the Earth (1); Classification makes the study of the large variety easy (1).	1+1
12	Heterotrophic, cell wall of chitin (1); Saprophytes; e.g. yeast and mushrooms (1).	1+1
13	Platyhelminthes are flat, nematodes are cylindrical (1); Examples: tapeworm and Ascaris (1).	1+1
14	Habitat and the form of organisms (1); Plants: herbs, shrubs, trees; animals by place of living (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cell type (prokaryotic/eukaryotic) (1); Body organisation (unicellular/multicellular) (1); Mode of nutrition and presence of cell wall (1).	1+1+1
16	Porifera: pores, canal system, sessile (1); Cnidaria: stinging cells, radial symmetry (1); Examples: Sycon; Hydra (1).	1+1+1
17	(a) Cnidaria; (b) Platyhelminthes (1); (c) Arthropoda (Insecta); (d) Amphibia (1); (e) Mammalia (1).	1+1+1
	OR: They feed on dead and decaying matter (1); Break it down and recycle nutrients (1); Saprophytic nutrition (1).	1+1+1
18	Segmented body (1); Coelom and bilateral symmetry (1); Earthworm, leech (1).	1+1+1
	OR: Kingdom Animalia, phylum Chordata (1); Class Mammalia, order Primates (1); Family Hominidae, genus Homo, species sapiens (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Monera (1); Protista (1); Fungi (1); Plantae (1); Animalia (1).	1+1+1+1+1
	OR: Cotyledons (1); Roots (1); Leaf venation (1); Flower parts (1); Examples (1).	1+1+1+1+1
20	Algae (1); Bryophyta (1); Pteridophyta (1); Gymnosperms (1); Angiosperms (1).	1+1+1+1+1
	OR: Arthropoda: jointed legs, exoskeleton (1); Mollusca: soft body, shell (1); Insects: pollination, silk, honey (1); Mollusca: food, pearls (1); Harm: pests, snails (1).	1+1+1+1+1

Mistakes to avoid:

- Writing scientific names wrongly: *homo Sapiens* or *Homo Sapiens*; the genus has a capital letter, the species a small letter, and both are italicised or underlined separately.
- Treating the kingdom as the smallest group and the species as the largest; the hierarchy runs from kingdom (largest) to species (smallest).
- Putting fungi (mushroom, yeast) in the plant kingdom; they lack chlorophyll and have chitin in their cell wall.
- Mixing up gymnosperms (naked seeds) and angiosperms (seeds enclosed in fruits), or monocots and dicots (one or two cotyledons).
- Saying that all vertebrates are warm-blooded, or that bats (mammals) are birds because they fly.
- Mixing up the phyla: spiders and insects are arthropods, earthworms are annelids, and roundworms are nematodes; also confusing 'Annelida' (segmented) with 'Nematoda' (unsegmented).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Platyhelminthes	Flat body.
2	**(b)** Reptilia	Scales, lay eggs.
3	**(b)** genus	First word is the genus.
4	**(c)** gymnosperms	e.g. pine.
5	**(a)** Nematoda	Cylindrical body.
6	**(c)** birds	Aves.
7	**(b)** kingdom, phylum, class, order, family, genus, species	Broad to specific.
8	**(b)** angiosperms	Flowering plants.
9	**(a)** Annelida	Segmented.
10	**(d)** mammary glands and hair	Feed young with milk.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They live on land but need water for fertilisation (1); No vascular tissue (1).	1+1
12	Animals with a notochord at some stage but no backbone (1); Balanoglossus, Herdmania (1).	1+1
13	Naming with two words: genus and species (1); Homo sapiens (1).	1+1
14	Gymnosperms: naked seeds, no fruits (1); Angiosperms: seeds enclosed in fruits, with flowers (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cotyledons: one vs two (1); Venation: parallel vs reticulate (1); Roots: fibrous vs tap root (1).	1+1+1
16	Soft body with a shell (1); Muscular foot (1); Snail, octopus (1).	1+1+1
17	Genus starts with a capital letter (1); Species starts with a small letter (1); Written in italics or underlined (1).	1+1+1
	OR: Unicellular prokaryotes (1); No nucleus or organelles (1); e.g. bacteria and blue-green algae (1).	1+1+1
18	Spiny skin (1); Radial symmetry (1); Starfish, sea urchin (1).	1+1+1
	OR: Useful: curd, nitrogen fixation (1); Harmful: cholera, typhoid (1); Role in decomposition (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Porifera (1); Cnidaria (1); Platyhelminthes and Nematoda (1); Annelida and Mollusca (1); Arthropoda and Echinodermata (1).	1+1+1+1+1
	OR: Fungi: yeast, mushroom (1); Bacteria: curd, cholera (1); Protozoa: malaria, amoebiasis (1); Decomposers (1); Uses and harm (1).	1+1+1+1+1
20	Pisces (1); Amphibia (1); Reptilia (1); Aves (1); Mammalia (1).	1+1+1+1+1
	OR: Vertebrates have a backbone (1); Invertebrates lack it (1); Cold-blooded: fish, amphibia, reptiles (1); Warm-blooded: birds, mammals (1); Body temperature (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** R. H. Whittaker	1969.
2	**(d)** monocots	e.g. wheat.
3	**(a)** Arthropoda	Jointed legs.
4	**(c)** platypus	Monotreme.
5	**(b)** Monera	Prokaryotes.
6	**(b)** dicot	Two cotyledons.
7	**(b)** Mollusca	Soft body with shell.
8	**(c)** birds and mammals	Constant body temperature.
9	**(c)** Protista	Unicellular eukaryotes.
10	**(d)** monocots	Grass and wheat.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Makes the study of the large variety of organisms easy (1); Shows relationships and helps in identification (1).	1+1
12	Monera, Protista, Fungi (1); Plantae and Animalia (1).	1+1
13	Jointed legs (1); Exoskeleton of chitin (1).	1+1
14	Feathers and beak (1); Warm-blooded, lay eggs, forelimbs are wings (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Unicellular eukaryotes (1); Plant-like, animal-like or fungus-like (1); Examples: Amoeba, Euglena (1).	1+1+1
16	Fish: gills, scales, aquatic (1); Amphibia: skin and lungs, moist skin (1); Reptiles: dry scaly skin, land, lay eggs (1).	1+1+1
17	Streamlined body and fins (1); Gills in fish for breathing in water (1); Moist skin and lungs in amphibians (1).	1+1+1
	OR: Algae: Spirogyra (1); Bryophytes: moss; pteridophytes: fern (1); Gymnosperms: pine; angiosperms: mango (1).	1+1+1
18	Birds: feathers, lay eggs (1); Mammals: hair, mammary glands, give birth (1); Both warm-blooded (1).	1+1+1
	OR: Bacteria, Archaea, Eukarya (1); Based on cell structure and genetics (1); Higher level than kingdom (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Aristotle: habitat-based (1); Linnaeus: binomial system and hierarchy (1); Whittaker: five kingdoms (1); Basis of each (1); Improvements over time (1).	1+1+1+1+1
	OR: Moss: spores, needs water (1); Fern: spores, vascular tissue (1); Pine: cones with naked seeds (1); Alternation of generations (1); Habitat (1).	1+1+1+1+1
20	Kingdom to species list (1); Mango: Plantae, Angiosperm, Dicot... (1); Dog: Animalia, Chordata, Mammalia, Carnivora, Canidae (1); Genus and species names (1); Common groups share features (1).	1+1+1+1+1
	OR: Universal names (1); Avoid confusion with local names (1); Mango, lion, human (1); Genus and species (1); Rules of nomenclature (1).	1+1+1+1+1

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